

Extended abstract prepared for hEART2014

Considering Overtaking and Common Lines in the Bus Bunching Problem

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$$t_{mn} = m\tau + n(k\tau + T) + \alpha \frac{(n+m-2)!}{(n-1)!(m-1)!} \left[\frac{k}{k-1} \right]^{m-1} \left[\frac{1}{1-k} \right]^{n-1} \quad (1)$$

$$p_{nm(i)}^s = \int_0^\infty f_i(\tau_{nm(i)}) \prod_{j \in L_n^s, m(j) \neq m(i)} \text{Prob} \left(E(\tau_{nm(j)}) > \tau + E(g_{nm(i)}^s) - E(g_{nm(j)}^s) \right) d\tau \quad (2)$$

$$k_{nm(i)} = \frac{\sum_s A_n^s p_{nm(i)}^s}{B_{m(i)}} \quad (3)$$

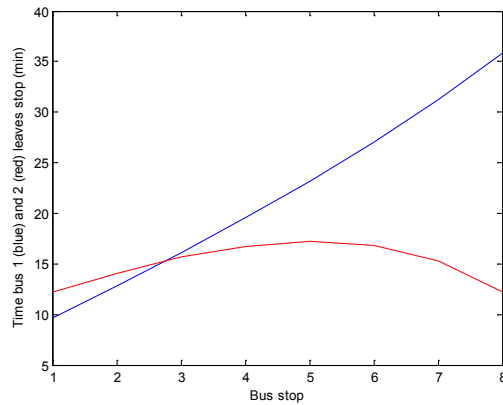


Fig. 1 Time of bus 1 (blue) and bus 2 (red) traversed through bus stops.

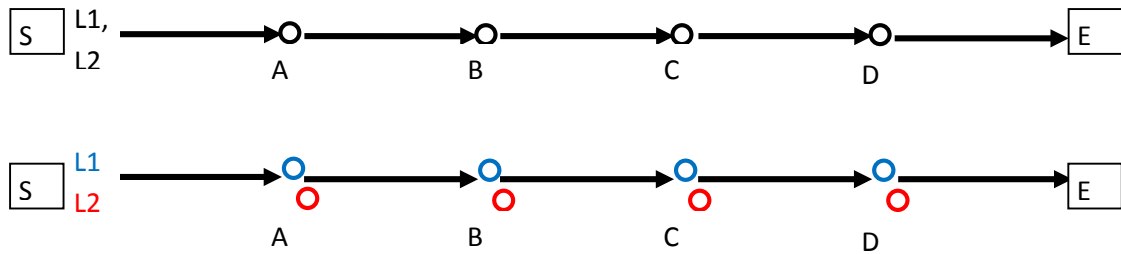


Fig. 2. Top: Two bus lines sharing the same stops, bottom: separate stops for both bus lines.

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References

The archived submission did not contain a separately labelled references section.